

Deleting a bracketed pair preserves length-additivity (and the bracket is not needed)

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This note is new at the commit named above; it depends on that commit only through [1], whose Lemmas 2.2 and 2.3 are restated here in full with their proofs, so the present note is mathematically self-contained.

Abstract

Let $S, T \subsetneq \mathbb{Z}/n$ and suppose the factorisation $u_S u_T$ of cyclically decreasing elements of the affine symmetric group is length-additive. Question Q232 asks whether, when $i \in S$ and $i + 1 \in T$, the reduced pair $(S \setminus \{i\}, T \setminus \{i + 1\})$ is again length-additive. We prove that it is. In fact we prove more: the hypothesis $i + 1 \in T$ is never used, so for *every* $i \in S$ the pair $(S \setminus \{i\}, T \setminus \{i + 1\})$ is length-additive, whether or not $i + 1$ lies in T . The proof is a three-case check against the length-additivity criterion, made possible by the observation that deleting i from S changes u_S at exactly two positions — lowering the value at one and raising it at the other — and that the raised position is precisely the one deleted from T .

1 Conventions, and the two facts we import

Let $n \geq 2$. The affine symmetric group $\tilde{S}_n$ is the group of bijections $w : \mathbb{Z} \rightarrow \mathbb{Z}$ with $w(j + n) = w(j) + n$ and $\sum_{j=1}^n w(j) = \binom{n+1}{2}$; for $i \in \mathbb{Z}/n$, s_i interchanges $i + jn$ and $i + 1 + jn$ for all j . Products act by $(ab)(j) = a(b(j))$ and $\ell(ws_i) = \ell(w) + 1$ iff $w(i) < w(i + 1)$. Every $w \in \tilde{S}_n$ is n -equivariant:

$$w(x + n) = w(x) + n \quad (x \in \mathbb{Z}). \quad (1)$$

A subset $S \subseteq \mathbb{Z}/n$ is identified with its n -periodic lift $\{j \in \mathbb{Z} : j \bmod n \in S\}$, written again S . A *run* of S is a maximal integer interval $[m, M] \subseteq S$; a *gap* is an element of $\mathbb{Z} \setminus S$. If $S \subsetneq \mathbb{Z}/n$ then the gap set is nonempty and n -periodic, so gaps recur with period n and

$$\text{every run of a proper } S \text{ has length } \leq n - 1. \quad (2)$$

The *blocks* of S are the intervals $[g + 1, g']$ for consecutive gaps $g < g'$; they partition $\mathbb{Z}$, and a run $[m, M]$ gives the block $[m, M + 1]$. For $S \subsetneq \mathbb{Z}/n$ we write u_S for the unique cyclically decreasing element with support S . We call (S, T) *additive* if $S, T \subsetneq \mathbb{Z}/n$ and $\ell(u_S u_T) = |S| + |T|$.

We import two facts from [1, Lems. 2.2, 2.3], restated here with their proofs.

Lemma 1.1 (Block model, [1, Lem. 2.2]). *Let $S \subsetneq \mathbb{Z}/n$. Then u_S rotates each block downwards by one step:*

$$u_S(g + 1) = g', \quad u_S(j) = j - 1 \quad (g + 1 < j \leq g') \quad (3)$$

for every block $[g+1, g']$. Equivalently $u_S(j) = j-1$ if $j-1 \in S$, and $u_S(j) = \min\{c \geq j : c \notin S\}$ otherwise. Moreover $\ell(u_S) = |S|$.

Proof. For a run $[m, M]$ the product $s_M s_{M-1} \cdots s_m$ is a reduced word in distinct generators in which s_{i+1} precedes s_i whenever both occur; as a permutation it sends $m \mapsto M+1$ and $j \mapsto j-1$ for $m < j \leq M+1$, i.e. it is the rotation (3) of the block $[m, M+1]$ and the identity elsewhere. Distinct runs are separated by a gap, so their blocks are disjoint and the rotations commute; the product is the stated permutation, of length $\sum (M-m+1) = |S|$. See [1, Lem. 2.2] for reducedness and uniqueness, which we do not use below. $\square$

Lemma 1.2 (Length-additivity criterion, [1, Lem. 2.3]). *Let $w \in \tilde{S}_n$ and $X \subsetneq \mathbb{Z}/n$. Then $\ell(wu_X) = \ell(w) + |X|$ if and only if for every run $[m, M]$ of X ,*

$$w(M+1) > w(j) \quad \text{for all } j \in [m, M].$$

Proof. For a single run, $u_{[m, M]} = s_M s_{M-1} \cdots s_m$. Multiplying on the right one letter at a time, $\ell(ws_M) = \ell(w) + 1$ iff $w(M) < w(M+1)$; the element ws_M has $M+1$ and M swapped in position, so $\ell(ws_M s_{M-1})$ increases iff $(ws_M)(M-1) < (ws_M)(M)$, i.e. iff $w(M-1) < w(M+1)$; inductively the j -th step increases the length iff $w(M-j+1) < w(M+1)$. Hence the run contributes $M-m+1$ iff $w(M+1) > w(j)$ for all $j \in [m, M]$. Distinct runs act on disjoint position-blocks and commute, and each condition involves only w , so the conditions are simultaneous. By (1) the condition for a run is invariant under replacing $[m, M]$ by $[m+kn, M+kn]$, so it suffices to check one lift of each run. $\square$

2 Deleting one element from S

Fix $S \subsetneq \mathbb{Z}/n$ and $i \in S$; choose an integer lift of i , again written i , and let $[p, q]$ be the run of S containing i , so

$$p \leq i \leq q, \quad p-1 \notin S, \quad q+1 \notin S, \quad q-p \leq n-2 \quad (4)$$

by (2). Write $S' = S \setminus \{i\}$ throughout this section.

Lemma 2.1 (Two-position perturbation). *The maps u_S and $u_{S'}$ agree at every $x \in \mathbb{Z}$ with $x \not\equiv p$ and $x \not\equiv i+1 \pmod{n}$, and*

$$u_S(p) = q+1, \quad u_{S'}(p) = i, \quad u_S(i+1) = i, \quad u_{S'}(i+1) = q+1.$$

In particular $u_{S'}(p) < u_S(p)$ and $u_{S'}(i+1) > u_S(i+1)$.

Proof. The gap set of S' is that of S together with the residue class of i . Hence every block of S other than $[p, q+1]$ and its translates is a block of S' , while $[p, q+1]$ splits into the two blocks $[p, i]$ and $[i+1, q+1]$ (both nonempty, as $p \leq i$ and $i+1 \leq q+1$). Outside the translates of $[p, q+1]$ the two elements are therefore given by the same rotations and agree. Inside, Lemma 1.1 gives

$$\begin{aligned} u_S : p &\mapsto q+1, & j &\mapsto j-1 \quad (p < j \leq q+1), \\ u_{S'} : p &\mapsto i, & j &\mapsto j-1 \quad (p < j \leq i), & i+1 &\mapsto q+1, & j &\mapsto j-1 \quad (i+1 < j \leq q+1). \end{aligned}$$

These agree at every $j \in [p, q+1]$ except $j = p$ and $j = i+1$, which are distinct because $p \leq i < i+1$. The displayed values are read off directly, and $u_{S'}(p) = i \leq q < q+1 = u_S(p)$, $u_{S'}(i+1) = q+1 > i = u_S(i+1)$. $\square$

Lemma 2.2 (The two moved positions are incongruent). $p \not\equiv i + 1 \pmod{n}$.

Proof. By (4), $i - p \leq q - p \leq n - 2$, so $i + 1 - n \leq p - 1 < p$; and $p \leq i < i + 1$. Hence $i + 1 - n < p < i + 1$, an open interval of length n whose only points congruent to $i + 1$ modulo n are its endpoints. So $p \not\equiv i + 1$. $\square$

Lemma 2.3 (Nothing below p climbs above $p - 1$). *For every $x \leq p - 1$ we have $u_S(x) \leq p - 1$ and $u_{S'}(x) \leq p - 1$.*

Proof. By (4), $p - 1$ is a gap of S , hence also of $S' \subseteq S$. Let $[g + 1, g']$ be the block of S (resp. of S') containing x , so $g < x \leq g'$ with g, g' consecutive gaps. If $g' > p - 1$ then $g < x \leq p - 1 < g'$, so the gap $p - 1$ lies strictly between the consecutive gaps g and g' — impossible. Hence $g' \leq p - 1$. By Lemma 1.1 the block is preserved, so $u(x) \in [g + 1, g']$ and $u(x) \leq g' \leq p - 1$. $\square$

3 The theorem

Theorem 3.1 (Deletion). *Let $n \geq 2$, let (S, T) be additive, and let $i \in S$. Then*

$$\ell(u_{S \setminus \{i\}} u_{T \setminus \{i+1\}}) = |S \setminus \{i\}| + |T \setminus \{i+1\}|,$$

i.e. $(S \setminus \{i\}, T \setminus \{i+1\})$ is additive.

Proof. Write $S' = S \setminus \{i\}$ and $T' = T \setminus \{i+1\}$, and fix an integer lift i of i with its S -run $[p, q]$ as in §2. Both S' and T' are proper subsets of $\mathbb{Z}/n$, and $\ell(u_{S'}) = |S'|$ by Lemma 1.1. By Lemma 1.2 applied with $w = u_{S'}$ and $X = T'$, it suffices to prove

$$u_{S'}(M' + 1) > u_{S'}(j) \quad \text{for every run } [m', M'] \text{ of } T' \text{ and every } j \in [m', M']. \quad (5)$$

If $T' = \emptyset$ there is nothing to prove. Fix a run $[m', M']$ of T' ; by (1) we may translate it by any multiple of n , since (5) is invariant under that translation. By (2),

$$m' \geq M' - n + 2. \quad (6)$$

Two remarks are used repeatedly. First, every $j \in [m', M']$ lies in T' , so

$$j \not\equiv i + 1 \pmod{n} \quad (j \in [m', M']), \quad (7)$$

because T' is the periodic lift of a set not containing the residue of $i + 1$. Second, by Lemma 2.1 and (7),

$$u_{S'}(j) \leq u_S(j) \quad (j \in [m', M']), \quad (8)$$

with equality unless $j \equiv p$.

We distinguish cases according to the residue of $M' + 1$. By Lemma 2.2 the two conditions $M' + 1 \equiv p$ and $M' + 1 \equiv i + 1$ are mutually exclusive.

Case 2: $M' + 1 \equiv p \pmod{n}$. Translate the run so that $M' + 1 = p$, i.e. $M' = p - 1$. Then $u_{S'}(M' + 1) = u_{S'}(p) = i$ by Lemma 2.1. Every $j \in [m', M']$ satisfies $j \leq p - 1$, so $u_{S'}(j) \leq p - 1$ by Lemma 2.3. Since $p \leq i$ we get $u_{S'}(j) \leq p - 1 \leq i - 1 < i = u_{S'}(M' + 1)$, which is (5).

In the remaining cases $M' + 1 \not\equiv p$, so Lemma 2.1 gives

$$u_{S'}(M' + 1) \geq u_S(M' + 1). \quad (9)$$

Since $M' + 1 \notin T'$, either $M' + 1 \notin T$ or else $M' + 1 \in T$ and $M' + 1 \equiv i + 1$.

Case 1a: $M' + 1 \notin T$. Then M' is the top of the run $[a, b]$ of T containing $[m', M']$ (such a run exists because $T' \subseteq T$ and $[m', M']$ is an interval in T'), i.e. $b = M'$. Additivity of (S, T) and Lemma 1.2 applied with $w = u_S$ and $X = T$ give $u_S(b + 1) > u_S(j)$ for all $j \in [a, b]$, hence in particular

$$u_S(M' + 1) > u_S(j) \quad (j \in [m', M']).$$

Combining with (9) on the left and (8) on the right, $u_{S'}(M' + 1) \geq u_S(M' + 1) > u_S(j) \geq u_{S'}(j)$, which is (5).

Case 1b: $M' + 1 \in T$ and $M' + 1 \equiv i + 1 \pmod{n}$. Translate the run so that $M' + 1 = i + 1$, i.e. $M' = i$. Then $u_{S'}(M' + 1) = u_{S'}(i + 1) = q + 1$ by Lemma 2.1, and by (6) we have $m' \geq i - n + 2$. Let $j \in [m', i]$; we show $u_{S'}(j) < q + 1$, which is (5).

If $j \equiv p \pmod{n}$. The points of $[m', i]$ congruent to p are among $p - n, p, p + n$. Now $p + n > i$ because $p \geq i - n + 2$, and $p - n < m'$ because $p - n \geq m'$ would force $p \geq m' + n \geq i + 2$, contradicting $p \leq i$. So $j = p$, and $u_{S'}(p) = i \leq q < q + 1$.

If $j \not\equiv p \pmod{n}$. Then $u_{S'}(j) = u_S(j)$ by Lemma 2.1 together with (7). If $p < j \leq i$ then $j - 1 \in [p, i - 1] \subseteq S$, so $u_S(j) = j - 1 \leq i - 1 < q + 1$ by Lemma 1.1. If $j < p$ then $u_S(j) \leq p - 1 < q + 1$ by Lemma 2.3. (The case $j > i$ does not occur, as $j \leq M' = i$; and $j = p$ was treated above.)

The three cases are exhaustive, so (5) holds for every run of T' , and Lemma 1.2 gives $\ell(u_{S'}u_{T'}) = \ell(u_{S'}) + |T'| = |S'| + |T'|$. $\square$

Corollary 3.2 (Q232, bracketed-pair deletion). *Let $n \geq 3$, let (S, T) be additive, and suppose $i \in S$ and $i + 1 \in T$. Then*

$$\ell(u_{S \setminus \{i\}} u_{T \setminus \{i+1\}}) = |S| + |T| - 2.$$

Proof. Theorem 3.1 with $|S \setminus \{i\}| = |S| - 1$ (as $i \in S$) and $|T \setminus \{i + 1\}| = |T| - 1$ (as $i + 1 \in T$). $\square$

4 The bracket hypothesis is not load-bearing

The brief for this session asserted that the hypothesis $i + 1 \in T$ must be consumed by any correct proof, on the ground that the *unbracketed analogue* — additivity is hereditary under $S' \subseteq S$, $T' \subseteq T$ — is false, with smallest witness $n = 3$, $S = \{0\}$, $T = \{0, 1\}$, $S' = T' = \{0\}$. That inference does not go through, and it is worth saying exactly why, because the conclusion drawn from it (“any proof that does not use the bracket is wrong”) would have rejected the proof above.

The false analogue and Theorem 3.1 differ in *which* elements are deleted, not in how many. In the witness $S = \{0\}$, $T = \{0, 1\}$, $S' = T' = \{0\}$, nothing is deleted from S at all; the deletion is $T \mapsto T \setminus \{1\}$ with S held fixed. A second witness from the same census, $n = 3$, $S = \{0, 1\}$, $T = \{1\}$, $S' = \{1\}$, $T' = \{1\}$, is the mirror failure: here $i = 0 \in S$ is deleted from S and $i + 1 = 1 \in T$ is *not* deleted from T , and $(\{1\}, \{1\})$ is indeed not additive ($u_{\{1\}}u_{\{1\}} = s_1s_1 = \text{id}$, of length $0 \neq 2$). What these show is that the two deletions must be *coupled*: removing i from S obliges one to remove $i + 1$ from T if it is there. They say nothing about whether $i + 1$ must be in T to begin with.

Theorem 3.1 is exactly this coupled statement, and the coupling is what the proof consumes, at (7): deleting i from S *raises* the value of u_S at the single position $i + 1$ (Lemma 2.1), which is the only way the criterion of Lemma 1.2 could be broken on the right-hand side, and that position

is unavailable as a j precisely because $i + 1$ was deleted from T . So the bracket appears in the proof as the *form of the conclusion*, not as a hypothesis: it is the reason T' is $T \setminus \{i + 1\}$ rather than some other subset. When $i + 1 \notin T$ the deletion is empty, $T' = T$, and the same argument shows $(S \setminus \{i\}, T)$ is additive. A census confirms this is not vacuous: over $3 \leq n \leq 8$ there are 60 082 additive triples (S, T, i) with $i \in S$ and $i + 1 \notin T$, and $(S \setminus \{i\}, T)$ is additive in all of them (§5).

Remark 4.1. The structural reason the coupled statement is so much better behaved than the general hereditary one is visible in Lemma 2.1. Deleting a single element from S is a minimal perturbation of u_S : it moves exactly two values, and it moves them in opposite directions. The criterion of Lemma 1.2 is monotone — it wants the value at $M' + 1$ large and the values on $[m', M']$ small — so a *lowered* value is harmless at j and a *raised* value is harmless at $M' + 1$. Of the four position/direction combinations, three are therefore automatically benign, and the fourth (the raised value appearing as some j) is killed by the deletion from T . What remains is the single genuinely new case, Case 2, where the lowered value appears at $M' + 1$; and there Lemma 2.3 applies because the position in question is the bottom of a block, so everything below it is trapped below it.

5 Computational verification

All code is in `proofs/code-q232/` and `reviews/code-2026-09-22-c2/` of the commit named on the title page. Affine permutations are held in window notation and lengths are computed by Shi’s inversion formula, not by the block model, so Lemma 1.1 is tested rather than assumed.

The statement. `g1_lemma.py` re-run on 23 September 2026: over all additive pairs with $3 \leq n \leq 8$ and all i with $i \in S$, $i + 1 \in T$, the reduced pair is additive in 82 044 of 82 044 instances. `g1_lemma2.py` re-run the same day: the hereditary analogue fails in 412 578 of 1 822 537 instances for $3 \leq n \leq 7$. Both counts reproduce those recorded in [2, §Gaps, (G1)] exactly.

The strengthening. `code-q232/stronger.py`: over $3 \leq n \leq 8$, all additive (S, T) and all $i \in S$ with $i + 1 \notin T$, the pair $(S \setminus \{i\}, T)$ is additive in 60 082 of 60 082 instances, 0 failures.

The mechanism, not only the conclusion. `code-q232/mechanism.py` checks the individual structural assertions of the proof rather than its conclusion, over all additive (S, T) with $2 \leq n \leq 8$ and all $i \in S$ (142 130 instances):

- Lemma 2.1: $u_{S'}$ and u_S differ at x if and only if $x \equiv p$ or $x \equiv i + 1$, and the four displayed values are as stated — 142 130 of 142 130, no exception;
- Lemma 2.2: $p \neq i + 1$ — 142 130 of 142 130;
- Lemma 2.3: $u_S(x), u_{S'}(x) \leq p - 1$ for all $x \leq p - 1$ — 142 130 of 142 130;
- the case split is exhaustive and no branch is vacuous: every run of T' falls in exactly one of Cases 1a, 1b, 2, with 151 052, 38 192 and 64 920 occurrences respectively, and the assertion “Case 1b $\Rightarrow M' + 1 \equiv i + 1$ ” never failed;
- the criterion (5) holds on every run in every case (0 failures), and the case-specific reasons hold too: in Case 2, $u_{S'}(j) \leq p - 1$ throughout the run; in Case 1a, $u_{S'}(j) \leq u_S(j)$ on the run and $u_{S'}(M' + 1) \geq u_S(M' + 1)$;

- Lemma 1.2 agrees with Shi’s formula on all 142 130 reduced pairs and on every original pair, so the criterion used in the proof is the same predicate that the censuses measure.

6 Consequence, and what it does and does not close

Theorem and section numbers of [1, 2] quoted below are those of the build at the commit named on the title page; the internal labels, which are stable, are `lem:blocks` = Lem. 2.2 and `lem:add` = Lem. 2.3 and `thm:exchange` = Thm. 4.1 and `lem:apply` = Lem. 4.2 in [1], and `lem:step1` = Lem. 3.1 and `thm:contain` = Thm. 3.2 and `sec:emptyX` = §5 in [2].

Gap (G1) of [2] records [2, §5] (the case $\mathcal{X}(S, T) = \emptyset$) as computational rather than proved, and names the obstruction precisely: the proof of [2, Thm. 3.2(iii)] invokes [2, Lem. 3.1] (Step 1 of [1, Lem. 4.2]), which requires its pair to be additive, whereas the $\mathcal{X} = \emptyset$ construction of Morse–Schilling removes a bracketed pair $(b - 1, b)$ with $b - 1 \in S$ and $b \in T$ and proceeds with the *reduced* pair $(S \setminus \{b - 1\}, T \setminus \{b\})$.

Corollary 3.2 with $i = b - 1$ says exactly that the reduced pair is additive. Consequently:

- (a) [2, Lem. 3.1] is available for the reduced pair, and the named obstruction in (G1) is removed.
- (b) Moreover the reduced pair has $b \in \mathcal{X}(S \setminus \{b - 1\}, T \setminus \{b\})$: in the $\mathcal{X} = \emptyset$ regime $b \notin S$ by the Case-(3) block condition, so $b \notin S \setminus \{b - 1\}$, and $b \notin T \setminus \{b\}$ by construction. So [2, Thm. 3.2] applies verbatim to the reduced pair with $x = b$, and its conclusion holds there.

What remains. This is not yet the full upgrade of [2, §5] from computed to proved, and I want to be exact about the residue. [2, Thm. 3.2] applied to the reduced pair produces a Theorem-4.1 exchange move *for the reduced pair*. The claim of [2, §5] is a statement about moves on the *original* pair (S, T) , namely that each lies in $\mathcal{E}^{\text{mv}}(S, T)$. Transporting a move across the bracket removal — identifying the exchange move of $(S \setminus \{b - 1\}, T \setminus \{b\})$ with an exchange move of (S, T) — is a further step that Theorem 3.1 does not supply. I have not proved it; it holds in all 1636 instances with $n \leq 7$ per [2, §5]. I record it as the residual gap:

(G1') *Let (S, T) be additive with $S \cup T = \mathbb{Z}/n$, let b be a Case-(3) index, and let μ be the Theorem-4.1 exchange move furnished by [2, Thm. 3.2] for the reduced pair $(S \setminus \{b - 1\}, T \setminus \{b\})$ at $x = b$. Is μ , read back on (S, T) , an element of $\mathcal{E}^{\text{mv}}(S, T)$?*

With (G1') settled, [2, §5] upgrades to proved. Without it, what Theorem 3.1 buys is the removal of the additivity obstruction — which was the part of (G1) stated as not known, and the part that had blocked every route.

References

- [1] Clio, *An exchange move for cyclically decreasing factorisations, and the reduction of WZZ Problem 5.1 to a single hypothesis*, 21 September 2026, `clio-vega/proofs@22d1b22`.
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- [3] T. Lam, *Affine Stanley symmetric functions*, Amer. J. Math. 128 (2006), 1553–1586.
- [4] J. Morse and A. Schilling, *Crystal approach to affine Schubert calculus*, `arXiv:1408.0320`.